Basics of Counting

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Outline

1. Introduction
2. Product Rule
3. Sum Rule
4. Inclusion–Exclusion Principle
5. Division Rule
6. More Complex Counting Problems

Why Counting?

Motivation

Passwords, test cases, phone numbers, license plates, network addresses, and algorithm steps all raise the same question: *how many possibilities are there?*

Example teaser (passwords)

A password has length 6, 7, or 8, using digits or letters, and must include at least one digit. Counting techniques will let us compute the number of valid passwords efficiently and correctly.

Big picture

Counting methods underpin probability, algorithm analysis, cryptography, and more. We start with two core rules: **Product Rule** and **Sum Rule**.

Product Rule

Product Rule (multiplication)

If a procedure is a **sequence of independent tasks**:

If task 1 has n_1 ways and task 2 has n_2 ways (for each choice of task 1),

then the whole procedure has $n_1 \cdot n_2$ ways. More generally, for m consecutive tasks with $n_1, n_2, \ldots, n_m$ ways each, total $= \prod_{i=1}^{m} n_i$.

Assigning Offices

Problem

A company has two employees (Sanchez and Patel) and 12 distinct offices. In how many ways can we assign different offices to the two employees?

Solution

Task 1: assign an office to Sanchez $\rightarrow 12$ choices.
Task 2: assign a *different* office to Patel $\rightarrow 11$ choices.
By the product rule: $12 \cdot 11 = \boxed{132}$ ways.

Bit Strings of Length 7

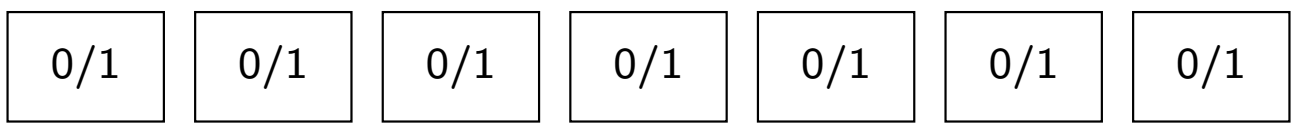
Problem

How many different bit strings of length 7 are there?

Solution

Each position has 2 choices (0 or 1), independently.

$$\text{Total} = 2^7 = \boxed{128}.$$



$$2 \times 2 \times \cdots \times 2 = 2^7$$

License Plates: LLL-DDD

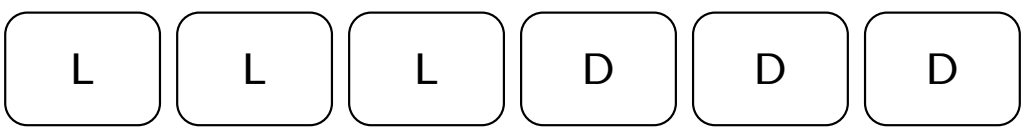
Problem

How many plates if the format is three uppercase letters followed by three digits? (Assume all sequences allowed.)

Solution

Each letter: 26 choices; each digit: 10 choices.

$$26^3 \cdot 10^3 = \boxed{17{,}576{,}000}.$$



$$26, 26, 26, 10, 10, 10$$

Nested Loops Count Iterations

Code pattern

```
k := 0
for i_1 := 1 to n_1
 for i_2 := 1 to n_2
  ⋮
 for i_m := 1 to n_m
  k := k + 1
```

Reasoning

The innermost statement executes for every combination of indices: $(i_1, i_2, \ldots, i_m)$ with n_1 choices for i_1, n_2 for i_2, …, n_m for i_m. By the product rule, the total is $\boxed{n_1 n_2 \cdots n_m}$. Hence final $k = n_1 n_2 \cdots n_m$.

Sum Rule (Addition Rule)

Sum Rule (addition)

If a procedure can be done by performing **one of several distinct tasks** — and no outcome is counted twice — then

$$\text{Total number of ways} = n_1 + n_2 + \cdots + n_m.$$

- Tasks are **mutually exclusive**: only one can be chosen.
- Compare with Product Rule: that applies when tasks are *independent and consecutive*.

Independent Loops: Sequential vs. Separate

Problem

What is the value of k after the following code, where $n_1, n_2, \ldots, n_m$ are positive integers?

```
k := 0
for i1 := 1 to n1
 k := k + 1
for i2 := 1 to n2
 k := k + 1
⋮
for im := 1 to nm
 k := k + 1
```

Solution

Each loop executes n_i times independently, one after another. Hence, the total count:

$$k = n_1 + n_2 + \cdots + n_m.$$

Choosing a Project from Different Lists

Problem

A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 projects respectively. No project appears on more than one list. How many possible projects are there to choose from?

Solution

Each list is a distinct, non-overlapping choice. By the sum rule:

$$23 + 15 + 19 = \boxed{57}.$$

Faculty or Student Representative

Problem

Suppose either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. There are 37 mathematics faculty and 83 mathematics majors. No one is both a faculty member and a student. How many different choices are there for the representative?

Solution

Two mutually exclusive options:

$$\text{Faculty: } 37 \quad \text{or} \quad \text{Student: } 83$$

By the sum rule:

$$37 + 83 = \boxed{120}.$$

Subtraction Rule (Inclusion–Exclusion Principle)

Definition

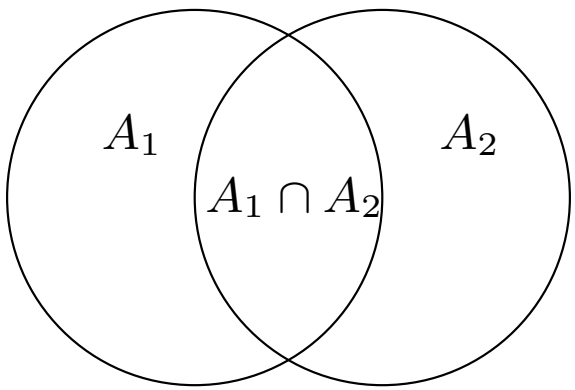
If a task can be done in either n_1 ways or n_2 ways, then the total number of ways to do the task is

$$n_1 + n_2 - (\text{ways common to both}).$$

Formula

For sets A_1, A_2:

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|.$$



- Use when two (or more) cases overlap.
- Subtract overlap to avoid double-counting.

Bit Strings with Overlapping Conditions

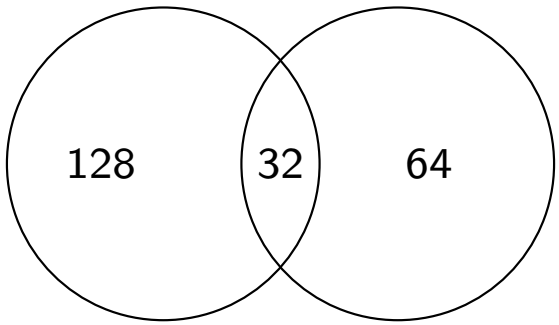
Problem

How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution

Let A_1: bit strings starting with $1 \to 2^7 = 128$ A_2: bit strings ending with $00 \to 2^6 = 64$ $A_1 \cap A_2$: start with 1 and end with $00 \to 2^5 = 32$

$$|A_1 \cup A_2| = 128 + 64 - 32 = \boxed{160}.$$



Total = 160

Applicants with Overlapping Majors

Problem

A company receives 350 applications. 220 majored in Computer Science, 147 majored in Business, and 51 majored in both. How many majored in neither?

Solution

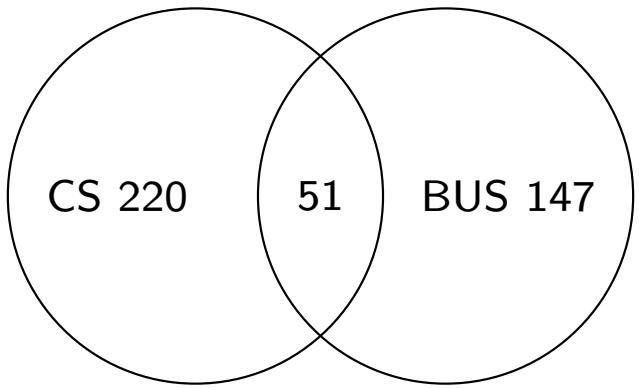
By inclusion–exclusion:

$$|CS \cup BUS| = 220 + 147 - 51 = 316.$$

Hence,

$$350 - 316 = \boxed{34}$$

majored in neither.



Division Rule

Definition

If a task can be done in n ways, but each distinct outcome is counted d times in that process, then the number of distinct ways is

$$\frac{n}{d}.$$

- Used when each outcome corresponds to multiple equivalent representations.
- Common in counting problems with **symmetry**.

Seating Around a Circular Table

Problem

How many different ways are there to seat four people around a circular table, where two seatings are considered the same if everyone has the same left and right neighbor?

Solution

There are $4!$ linear arrangements, but each circular seating corresponds to 4 rotations that look identical.

$$\text{Distinct seatings} = \frac{4!}{4} = \boxed{6}.$$

Practice 1: Divisibility by 7 and 11

Problem

How many positive integers less than 1000:

(a) are divisible by 7?

(b) are divisible by 7 but not by 11?

(c) are divisible by both 7 and 11?

(d) are divisible by either 7 or 11?

(e) are divisible by exactly one of 7 and 11?

(f) are divisible by neither 7 nor 11?

Practice 1: Divisibility by 7 and 11

Solution Outline

$$\lfloor 999/7 \rfloor = 142, \quad \lfloor 999/11 \rfloor = 90, \quad \lfloor 999/77 \rfloor = 12$$

- (a) 142
- (b) $142 - 12 = 130$
- (c) 12
- (d) $142 + 90 - 12 = 220$
- (e) $220 - 2 \times 12 = 196$
- (f) $999 - 220 = \boxed{779}$

Practice 2: Password Counting with Constraints

Problem

Each user on a computer system has a password that is 6–8 characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution

Each position: $36 = 26 + 10$ choices.

Total passwords:

$$36^6 + 36^7 + 36^8$$

Invalid (no digits):

$$26^6 + 26^7 + 26^8$$

Valid (at least one digit):

$$(36^6 - 26^6) + (36^7 - 26^7) + (36^8 - 26^8)$$

Practice 3: BASIC Variable Names

Problem

In a version of BASIC, variable names are 1 or 2 alphanumeric characters (letters or digits). Names must begin with a letter and cannot be any of 5 reserved 2-character strings. Uppercase and lowercase are not distinguished.

Solution

1-character names:

$$26$$

2-character names: First must be a letter (26), second can be letter or digit (36). Total $= 26 \times 36 = 936$.
Subtract reserved words: $936 - 5 = 931$.
Total:

$$26 + 931 = \boxed{957}$$

26 (1-char) + 931 (2-char) = 957